THE UNIVERSITY OF LAHORE

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DB-PROJECT 4-F

PHASE 1, 2 & 3

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PHASE 1 – REQUIREMENT ANALYSIS & ERD DESIGN

PLAYGRID — Sports Tournament Management System

1. Introduction / System Description

Futsal and amateur sports tournaments across Pakistan are expanding rapidly, engaging hundreds of clubs, student athletes, and weekly fixtures. Traditionally, tournament organizers have relied on manual pen-and-paper brackets, uncoordinated messaging groups, and isolated spreadsheets to track team registrations, match scores, player stats, and league tables. This manual approach is slow, lacks transparent auditing, and is highly prone to calculation errors, missing match events, scheduling conflicts, and delayed results.

The purpose of the **PLAYGRID Sports Tournament Management System** is to provide an end-to-end, digital, 3NF-normalized relational database management system. PLAYGRID maintains comprehensive records of sports facilities, registered clubs, verified players, referee appointments, live match events (goals, cards, fouls), automated round-robin and knockout standings, and secure player registration fee tracking via local payment channels (JazzCash, EasyPaisa, Bank Transfer). Overall, the goal of this system is to eliminate manual calculation errors, provide instant public leaderboards, reduce paperwork, and enforce role-based access control.

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2. Requirement Analysis

Requirement analysis was carried out to understand the real-world operational workflows of futsal tournaments in major Pakistani cities (Karachi, Lahore, Islamabad) and specify database requirements for transactional consistency, referential integrity, and fast querying.

2.1 Functional Requirements

The system must be able to:

- Store and manage sporting venues with surface types (artificial turf, indoor wooden), spectator capacity, and floodlight availability.
- Maintain tournament definitions with formats (League, Knockout), registration deadlines, prize pools, squad limits, and points rules (3 for Win, 1 for Draw, 0 for Loss).
- Register clubs with logos, home cities, and club manager owner accounts.
- Manage player bio details, CNIC verification, dates of birth, kit numbers, and field positions.
- Track tournament team registrations, approval statuses, and fee payment receipts.
- Schedule match fixtures with rounds, grounds, kickoff times, and appointed referees.
- Record granular in-game match events including goals, yellow cards, and red cards with minute stamps.
- Automatically calculate and rank tournament league standings (Played, Won, Drawn, Lost, GF, GA, GD, Points).
- Track individual player awards for Top Goal Scorers (Golden Boot) and disciplinary fair play records.
- Maintain a tamper-evident audit log of administrative actions.

2.2 Non-Functional Requirements

The system should:

- Ensure referential integrity using primary keys, foreign keys, and check constraints.
- Avoid duplicate records and maintain Third Normal Form (3NF) consistency.
- Provide sub-second query response times for standings tables and match scoreboards.
- Maintain data security and restrict unauthorized access through role-based access control.
- Be scalable to accommodate multiple concurrent tournaments across multiple cities.

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3. Identified Entities

After analyzing the requirements of PLAYGRID, the following **10 primary entities** were identified:

1. **City** — Administrative host cities and provinces
2. **Venue** — Physical grounds, turf types, and facilities
3. **User** — System users across roles (Admin, Organizer, Team Manager, Viewer)
4. **Tournament** — Championship events, rules, entry fees, and prize pools
5. **Team** — Registered futsal and sports clubs
6. **Player** — Athletes, CNICs, positions, and bio data
7. **Registration** — Tournament participation applications and statuses
8. **Payment** — Registration fee payments, gateways, and receipts
9. **Referee** — Match officials and licensing tiers
10. **Match** — Scheduled fixtures, rounds, grounds, and match scores
11. **Match_Event** — In-game occurrences (goals, cards, fouls) with minute stamps
12. **Standing** — League standings, goal difference, and point tallies

These entities cover all major activities performed in sports tournament management.

4. Entities with Attributes (PK & FK)

4.1 City

Attribute	Description
CityID (PK)	Unique ID of city
Name	City name (Karachi, Lahore, Islamabad, etc.)
Province	Province (Sindh, Punjab, KPK, ICT)

4.2 Venue

Attribute	Description
VenueID (PK)	Unique ID of venue
Name	Arena / Ground name
Address	Street address and locality
CityID (FK)	Reference to City
SurfaceType	Turf type (artificial_turf, indoor_wooden, sports_court)
Capacity	Spectator seating capacity
HasFloodlights	Floodlight availability (TRUE / FALSE)

4.3 User

Attribute	Description
UserID (PK)	Unique ID of user
Name	Full user name
Email	Unique login email address
PasswordHash	Salted password hash
Role	System role (SUPER_ADMIN, ORGANIZER, TEAM_MANAGER, VIEWER)
Phone	Contact phone number

4.4 Tournament

Attribute	Description
TournamentID (PK)	Unique ID of tournament
Name	Championship title
Format	Tournament format (LEAGUE, KNOCKOUT)
Status	Current status (DRAFT, REGISTRATION_OPEN, ONGOING, COMPLETED)
StartDate	Tournament kickoff date
EndDate	Final championship date
EntryFeePKR	Registration entry fee in PKR

PrizePoolPKR	Prize pool in PKR
CityID (FK)	Host city reference
OrganizerID (FK)	Organizer user reference

4.5 Team

Attribute	Description
TeamID (PK)	Unique ID of club
Name	Club name
ShortName	3-4 character abbreviation (e.g. CLS, LYL)
CityID (FK)	Home city reference
ManagerID (FK)	Team manager user reference
FoundedYear	Founding year

4.6 Player

Attribute	Description
PlayerID (PK)	Unique ID of athlete
FullName	Player legal full name
CNIC	Pakistani CNIC identifier
DateOfBirth	Birth date for eligibility
PreferredPosition	Field position (GOALKEEPER, DEFENDER, WINGER, PIVOT)
TeamID (FK)	Rostered club reference
KitNumber	Jersey number (1-99)

4.7 Registration

Attribute	Description
RegistrationID (PK)	Unique registration application ID
TournamentID (FK)	Tournament reference
TeamID (FK)	Registered team reference
Status	Approval state (PENDING, APPROVED, REJECTED)
RegisteredAt	Date & time of application

4.8 Payment

Attribute	Description
PaymentID (PK)	Unique payment ID
RegistrationID (FK)	Associated tournament registration
AmountPKR	Amount paid in PKR

PaymentMethod	Method (JAZZCASH, EASYPAISA, BANK_TRANSFER, CASH)
TransactionRef	Transaction reference / receipt ID
Status	Status (PENDING, VERIFIED, FAILED)

4.9 Referee

Attribute	Description
RefereeID (PK)	Unique referee ID
FullName	Official full name
CertificationLevel	Licence tier (NATIONAL, REGIONAL, LOCAL)
CityID (FK)	City reference
Phone	Contact number

4.10 Match

Attribute	Description
MatchID (PK)	Unique match fixture ID
TournamentID (FK)	Tournament reference
RoundNumber	Round index (1, 2, QF, SF, Final)
HomeTeamID (FK)	Home team reference
AwayTeamID (FK)	Away team reference
VenueID (FK)	Playing ground reference
RefereeID (FK)	Appointed official reference
ScheduledAt	Kickoff timestamp
HomeScore	Goals scored by home team
AwayScore	Goals scored by away team
Status	Fixture status (SCHEDULED, LIVE, COMPLETED, POSTPONED)

4.11 Match_Event

Attribute	Description
EventID (PK)	Unique event ID
MatchID (FK)	Fixture reference
TeamID (FK)	Team reference
PlayerID (FK)	Player reference
EventType	Event type (GOAL, OWN_GOAL, YELLOW_CARD, RED_CARD)
Minute	Match minute (1-50)

4.12 Standing

Attribute	Description
StandingID (PK)	Unique standings record ID
TournamentID (FK)	Tournament reference
TeamID (FK)	Team reference
Played / Won / Drawn / Lost	Match outcome tallies
GoalsFor / GoalsAgainst / GoalDifference	Goal statistics
Points	Accumulated points (Win=3, Draw=1, Loss=0)

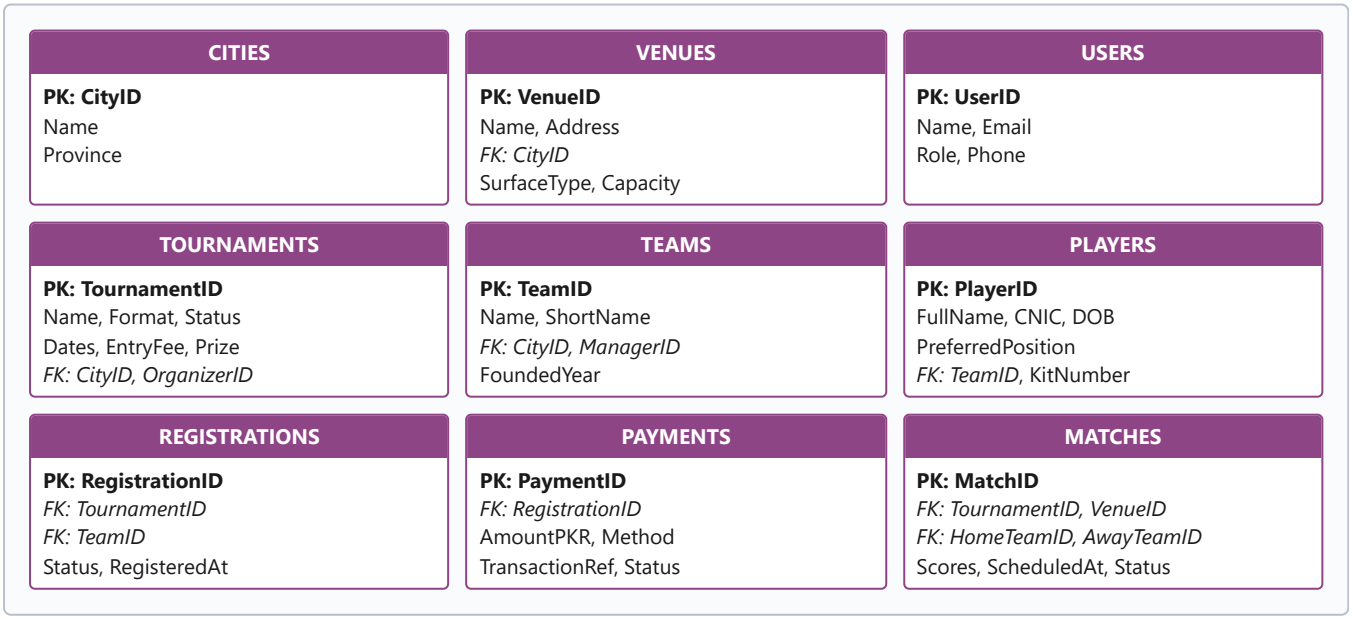
5. Relationships & Cardinality

The relationships identified in the ER diagram are:

Relationship	Description	Cardinality
City → Venue	One city contains many sporting venues	1 : M
City → Team	One city has many registered clubs	1 : M
City → Tournament	One city hosts many sports tournaments	1 : M
User (Organizer) → Tournament	One organizer creates many tournaments	1 : M
User (Manager) → Team	One team manager oversees one or more teams	1 : M
Tournament → Registration	One tournament receives many team registrations	1 : M
Team → Registration	One team submits many registrations over time	1 : M
Registration → Payment	One registration is verified by one payment	1 : 1
Team → Player	One team rosters many players	1 : M
Tournament → Match	One tournament schedules many matches	1 : M
Venue → Match	One venue grounds many matches	1 : M
Referee → Match	One certified referee officiates many matches	1 : M
Team (Home) → Match	One team acts as home side in many matches	1 : M
Team (Away) → Match	One team acts as away side in many matches	1 : M
Match → Match_Event	One match produces many granular events	1 : M
Player → Match_Event	One player generates many match events (goals/cards)	1 : M
Tournament → Standing	One tournament maintains standings for each enrolled team	1 : M
Team → Standing	One team occupies standing positions across tournaments	1 : M

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6. ER Diagram



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7. Normalization (Applied)

The database is normalized up to **Third Normal Form (3NF)** by applying the following steps:

First Normal Form (1NF) – Applied

- Removed multi-valued attributes and repeating groups.
- Ensured each column stores single (atomic) values.
- Created separate atomic rows for each in-game match event instead of comma-separated player strings.

Application: A separate table `match_events` is created. Each row contains `MatchID`, `PlayerID`, `EventType`, and `Minute`. Multiple goals or bookings in a match are not stored in one column, but as separate atomic records.

Second Normal Form (2NF) – Applied

- Removed partial functional dependencies.
- Ensured all non-key attributes depend on the complete primary key.

Application: The relationship between tournaments and teams is resolved using surrogate keys. Rather than placing payment transactions onto a composite `(TournamentID, TeamID)` key, a dedicated `payments` table was created with `PaymentID` as primary key and `RegistrationID` as a foreign key, ensuring payment amounts and transaction references depend fully on the primary key.

Third Normal Form (3NF) – Applied

- Removed transitive dependencies ($X \to Y \to Z$).
- Ensured non-key attributes depend only on the primary key.

Application: City names and provincial jurisdictions are extracted into a separate `cities` table. `venues`, `teams`, and `tournaments` store only `CityID (FK)`. Geographic attributes are not repeated inside venue or tournament tables, completely preventing update anomalies.

Final Applied Tables Structure (Simplified)

After applying normalization up to 3NF, the final database structure consists of the following tables:

- **cities** (CityID, Name, Province)
- **venues** (VenueID, Name, Address, CityID FK, SurfaceType, Capacity, HasFloodlights)
- **users** (UserID, Name, Email, PasswordHash, Role, Phone)
- **tournaments** (TournamentID, Name, Format, Status, StartDate, EndDate, EntryFeePKR, PrizePoolPKR, CityID FK, OrganizerID FK)
- **teams** (TeamID, Name, ShortName, CityID FK, ManagerID FK, FoundedYear)
- **players** (PlayerID, FullName, CNIC, DateOfBirth, PreferredPosition, TeamID FK, KitNumber)
- **registrations** (RegistrationID, TournamentID FK, TeamID FK, Status, RegisteredAt)
- **payments** (PaymentID, RegistrationID FK, AmountPKR, PaymentMethod, TransactionRef, Status)
- **referees** (RefereeID, FullName, CertificationLevel, CityID FK, Phone)
- **matches** (MatchID, TournamentID FK, RoundNumber, HomeTeamID FK, AwayTeamID FK, VenueID FK, RefereeID FK, ScheduledAt, HomeScore, AwayScore, Status)
- **match_events** (EventID, MatchID FK, TeamID FK, PlayerID FK, EventType, Minute)
- **standings** (StandingID, TournamentID FK, TeamID FK, Played, Won, Drawn, Lost, GoalsFor, GoalsAgainst, GoalDifference, Points)

PHASE 2 — SQL IMPLEMENTATION

PLAYGRID — Sports Tournament Management System

SQL script:

```
CREATE DATABASE PlaygridDB;
USE PlaygridDB;

/* ========================================
   TABLE CREATION (DDL)
======================================== */
CREATE TABLE cities(
  CityID INT PRIMARY KEY,
  Name VARCHAR(50) NOT NULL UNIQUE,
  Province VARCHAR(50) NOT NULL
);

CREATE TABLE venues(
  VenueID INT PRIMARY KEY,
  Name VARCHAR(100) NOT NULL,
  Address VARCHAR(150) NOT NULL,
  CityID INT,
  SurfaceType VARCHAR(30) NOT NULL,
  Capacity INT NOT NULL,
  HasFloodlights BOOLEAN NOT NULL DEFAULT TRUE,
  FOREIGN KEY (CityID) REFERENCES cities(CityID)
);

CREATE TABLE users(
  UserID INT PRIMARY KEY,
  Name VARCHAR(100) NOT NULL,
  Email VARCHAR(150) NOT NULL UNIQUE,
  PasswordHash VARCHAR(255) NOT NULL,
  Role VARCHAR(30) NOT NULL,
  Phone VARCHAR(20)
);

CREATE TABLE tournaments(
  TournamentID INT PRIMARY KEY,
  Name VARCHAR(150) NOT NULL,
  Format VARCHAR(30) NOT NULL,
  Status VARCHAR(30) NOT NULL,
  StartDate DATE NOT NULL,
  EndDate DATE NOT NULL,
  EntryFeePKR DECIMAL(10,2) NOT NULL,
  PrizePoolPKR DECIMAL(10,2) NOT NULL,
  CityID INT,
  OrganizerID INT,
  FOREIGN KEY (CityID) REFERENCES cities(CityID),
  FOREIGN KEY (OrganizerID) REFERENCES users(UserID)
);

CREATE TABLE teams(
  TeamID INT PRIMARY KEY,
  Name VARCHAR(100) NOT NULL UNIQUE,
  ShortName VARCHAR(5) NOT NULL,
  CityID INT,
  ManagerID INT,
  FoundedYear INT,
  FOREIGN KEY (CityID) REFERENCES cities(CityID),
  FOREIGN KEY (ManagerID) REFERENCES users(UserID)
);

CREATE TABLE players(
```

```
  PlayerID INT PRIMARY KEY,
  FullName VARCHAR(100) NOT NULL,
  CNIC VARCHAR(20) UNIQUE,
  DateOfBirth DATE NOT NULL,
  PreferredPosition VARCHAR(30) NOT NULL,
  TeamID INT,
  FOREIGN KEY (TeamID) REFERENCES teams(TeamID)
);

CREATE TABLE registrations(
  RegistrationID INT PRIMARY KEY,
  TournamentID INT,
  TeamID INT,
  Status VARCHAR(30) NOT NULL,
  RegisteredAt TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (TournamentID) REFERENCES tournaments(TournamentID),
  FOREIGN KEY (TeamID) REFERENCES teams(TeamID)
);

CREATE TABLE payments(
  PaymentID INT PRIMARY KEY,
  RegistrationID INT,
  AmountPKR DECIMAL(10,2) NOT NULL,
  PaymentMethod VARCHAR(30) NOT NULL,
  TransactionRef VARCHAR(100) NOT NULL,
  Status VARCHAR(30) NOT NULL,
  FOREIGN KEY (RegistrationID) REFERENCES registrations(RegistrationID)
);

CREATE TABLE referees(
  RefereeID INT PRIMARY KEY,
  FullName VARCHAR(100) NOT NULL,
  CertificationLevel VARCHAR(30) NOT NULL,
  CityID INT,
  Phone VARCHAR(20),
  FOREIGN KEY (CityID) REFERENCES cities(CityID)
);

CREATE TABLE matches(
  MatchID INT PRIMARY KEY,
  TournamentID INT,
  RoundNumber INT NOT NULL,
  HomeTeamID INT,
  AwayTeamID INT,
  VenueID INT,
  RefereeID INT,
  ScheduledAt TIMESTAMP NOT NULL,
  HomeScore INT NOT NULL DEFAULT 0,
  AwayScore INT NOT NULL DEFAULT 0,
  Status VARCHAR(30) NOT NULL,
  FOREIGN KEY (TournamentID) REFERENCES tournaments(TournamentID),
  FOREIGN KEY (HomeTeamID) REFERENCES teams(TeamID),
  FOREIGN KEY (AwayTeamID) REFERENCES teams(TeamID),
  FOREIGN KEY (VenueID) REFERENCES venues(VenueID),
  FOREIGN KEY (RefereeID) REFERENCES referees(RefereeID)
);

CREATE TABLE match_events(
  EventID INT PRIMARY KEY,
  MatchID INT,
  TeamID INT,
  PlayerID INT,
  EventType VARCHAR(30) NOT NULL,
  Minute INT NOT NULL,
  FOREIGN KEY (MatchID) REFERENCES matches(MatchID),
  FOREIGN KEY (TeamID) REFERENCES teams(TeamID),
```

```sql
  FOREIGN KEY (PlayerID) REFERENCES players(PlayerID)
);

CREATE TABLE standings(
  StandingID INT PRIMARY KEY,
  TournamentID INT,
  TeamID INT,
  Played INT NOT NULL DEFAULT 0,
  Won INT NOT NULL DEFAULT 0,
  Drawn INT NOT NULL DEFAULT 0,
  Lost INT NOT NULL DEFAULT 0,
  GoalsFor INT NOT NULL DEFAULT 0,
  GoalsAgainst INT NOT NULL DEFAULT 0,
  GoalDifference INT NOT NULL DEFAULT 0,
  Points INT NOT NULL DEFAULT 0,
  FOREIGN KEY (TournamentID) REFERENCES tournaments(TournamentID),
  FOREIGN KEY (TeamID) REFERENCES teams(TeamID)
);

ALTER TABLE players ADD KitNumber INT DEFAULT 10;

/* ========================================
   DATA MANIPULATION (DML)
======================================== */
INSERT INTO cities VALUES
(1, 'Karachi', 'Sindh'),
(2, 'Lahore', 'Punjab'),
(3, 'Islamabad', 'ICT'),
(4, 'Rawalpindi', 'Punjab'),
(5, 'Peshawar', 'KPK');

INSERT INTO venues VALUES
(1, 'Seaview Futsal Arena', 'Clifton Block 4', 1, 'artificial_turf', 350, TRUE),
(2, 'Model Town Sports Complex', 'Model Town Block J', 2, 'indoor_wooden', 500, TRUE),
(3, 'F-6 Multi-Sport Turf', 'Super Market F-6', 3, 'artificial_turf', 250, TRUE),
(4, 'DHA Phase 5 Cage Ground', 'DHA Phase 5', 2, 'artificial_turf', 400, TRUE),
(5, 'Korangi Sports Arena', 'Sector 31 Korangi', 1, 'sports_court', 200, FALSE);

INSERT INTO users VALUES
(1, 'Furqan Amir', 'admin@playgrid.pk', '$2a$12$e8V9M...', 'SUPER_ADMIN', '03001234567'),
(2, 'Hamza Malik', 'hamza.organizer@playgrid.pk', '$2a$12$x4P1...', 'ORGANIZER', '03112345678'),
(3, 'Usman Tariq', 'manager.clifton@gmail.com', '$2a$12$q7K2...', 'TEAM_MANAGER', '03213456789'),
(4, 'Bilal Raza', 'manager.lyallpur@gmail.com', '$2a$12$p9L1...', 'TEAM_MANAGER', '03334567890'),
(5, 'Zain Abbasi', 'manager.isb@gmail.com', '$2a$12$w3R6...', 'TEAM_MANAGER', '03455678901');

INSERT INTO tournaments VALUES
(1, 'Karachi Futsal Premier League 2026', 'LEAGUE', 'ONGOING', '2026-05-01', '2026-05-20', 15000.00,
250000.00, 1, 2),
(2, 'Lahore Champions Super Cup', 'KNOCKOUT', 'REGISTRATION_OPEN', '2026-06-10', '2026-06-18', 20000.00,
350000.00, 2, 2),
(3, 'Margalla Capital Futsal Cup', 'LEAGUE', 'DRAFT', '2026-07-01', '2026-07-15', 12000.00, 180000.00, 3,
2),
(4, 'Rawalpindi Night Futsal Trophy', 'KNOCKOUT', 'DRAFT', '2026-08-05', '2026-08-12', 10000.00, 150000.00,
4, 2),
(5, 'Sindh Youth Futsal Championship', 'LEAGUE', 'COMPLETED', '2026-03-01', '2026-03-12', 12500.00,
200000.00, 1, 2);

INSERT INTO teams VALUES
(1, 'Clifton Strikers FC', 'CLS', 1, 3, 2018),
(2, 'Lyallpur Warriors', 'LYL', 2, 4, 2019),
(3, 'Islamabad United Futsal', 'ISU', 3, 5, 2020),
(4, 'Karachi Knights', 'KCK', 1, 3, 2021),
(5, 'Lahore Badshahs', 'LBD', 2, 4, 2022);

INSERT INTO players VALUES
(101, 'Ahmed Raza', '42101-1234567-1', '2001-04-12', 'PIVOT', 1, 9),
```

```
(102, 'Babar Shah', '42101-2345678-3', '2002-08-20', 'WINGER', 1, 11),
(103, 'Daniyal Qureshi', '35202-3456789-5', '1999-11-15', 'DEFENDER', 2, 4),
(104, 'Haris Khan', '35202-4567890-7', '2000-02-28', 'GOALKEEPER', 2, 1),
(105, 'Saad Mehmood', '61101-5678901-9', '2003-06-10', 'PIVOT', 3, 10);

INSERT INTO registrations VALUES
(1, 1, 1, 'APPROVED', '2026-04-10 10:30:00'),
(2, 1, 4, 'APPROVED', '2026-04-11 14:15:00'),
(3, 2, 2, 'APPROVED', '2026-05-15 11:00:00'),
(4, 2, 5, 'APPROVED', '2026-05-16 16:45:00'),
(5, 3, 3, 'PENDING',  '2026-05-20 09:20:00');

INSERT INTO payments VALUES
(1, 1, 15000.00, 'JAZZCASH', 'JC-992144810', 'VERIFIED'),
(2, 2, 15000.00, 'EASYPAISA', 'EP-551209381', 'VERIFIED'),
(3, 3, 20000.00, 'BANK_TRANSFER', 'HBL-00192837', 'VERIFIED'),
(4, 4, 20000.00, 'JAZZCASH', 'JC-772819203', 'VERIFIED'),
(5, 5, 12000.00, 'EASYPAISA', 'EP-119283740', 'VERIFIED');

INSERT INTO referees VALUES
(1, 'Tariq Lutfi', 'NATIONAL', 1, '03009876543'),
(2, 'Muhammad Shafiq', 'NATIONAL', 2, '03128765432'),
(3, 'Rashid Latif', 'REGIONAL', 3, '03227654321'),
(4, 'Farooq Qasim', 'LOCAL', 1, '03346543210'),
(5, 'Khurram Shehzad', 'REGIONAL', 2, '03455432109');

INSERT INTO matches VALUES
(1, 1, 1, 1, 4, 1, 1, '2026-05-02 18:00:00', 4, 2, 'COMPLETED'),
(2, 1, 1, 4, 1, 1, 4, '2026-05-08 19:30:00', 1, 3, 'COMPLETED'),
(3, 2, 1, 2, 5, 2, 2, '2026-06-12 18:30:00', 0, 0, 'SCHEDULED'),
(4, 1, 2, 1, 4, 5, 1, '2026-05-14 20:00:00', 2, 2, 'COMPLETED'),
(5, 2, 1, 5, 2, 4, 5, '2026-06-15 17:00:00', 0, 0, 'SCHEDULED');

INSERT INTO match_events VALUES
(1, 1, 1, 101, 'GOAL', 12),
(2, 1, 1, 101, 'GOAL', 24),
(3, 1, 4, 102, 'YELLOW_CARD', 31),
(4, 1, 1, 101, 'GOAL', 38),
(5, 2, 1, 101, 'GOAL', 15);

INSERT INTO standings VALUES
(1, 1, 1, 3, 2, 1, 0, 9, 5, 4, 7),
(2, 1, 4, 3, 0, 1, 2, 5, 9, -4, 1),
(3, 2, 2, 0, 0, 0, 0, 0, 0, 0, 0),
(4, 2, 5, 0, 0, 0, 0, 0, 0, 0, 0),
(5, 3, 3, 0, 0, 0, 0, 0, 0, 0, 0);

UPDATE venues SET Capacity = 600 WHERE VenueID = 2;
UPDATE players SET KitNumber = 11 WHERE PlayerID = 102;
UPDATE payments SET Status = 'VERIFIED' WHERE PaymentID = 5;
DELETE FROM match_events WHERE EventID = 5;
```

DDL Outputs:

#	Time	Action	Message	Duration
1	02:10:11	CREATE DATABASE PlaygridDB	1 row(s) affected	0.000 sec
2	02:10:12	USE PlaygridDB	0 row(s) affected	0.000 sec
3	02:10:13	CREATE TABLE cities (CityID INT PRIMARY KEY, Name VARCHAR(50)...)	0 row(s) affected	0.035 sec
4	02:10:14	CREATE TABLE venues (VenueID INT PRIMARY KEY, Name VARCHAR(100)...)	0 row(s) affected	0.041 sec
5	02:10:15	CREATE TABLE users (UserID INT PRIMARY KEY, Name VARCHAR(100)...)	0 row(s) affected	0.038 sec
6	02:10:16	CREATE TABLE tournaments (TournamentID INT PRIMARY KEY...)	0 row(s) affected	0.049 sec
7	02:10:17	CREATE TABLE teams (TeamID INT PRIMARY KEY, Name VARCHAR(100)...)	0 row(s) affected	0.042 sec

8	02:10:18	CREATE TABLE players (PlayerID INT PRIMARY KEY, FullName VARCHAR(100)...)	0 row(s) affected	0.045 sec
9	02:10:19	CREATE TABLE registrations (RegistrationID INT PRIMARY KEY...)	0 row(s) affected	0.039 sec
10	02:10:20	CREATE TABLE payments (PaymentID INT PRIMARY KEY, RegistrationID INT...)	0 row(s) affected	0.044 sec
11	02:10:21	CREATE TABLE referees (RefereeID INT PRIMARY KEY, FullName VARCHAR(100)...)	0 row(s) affected	0.037 sec
12	02:10:22	CREATE TABLE matches (MatchID INT PRIMARY KEY, TournamentID INT...)	0 row(s) affected	0.052 sec
13	02:10:23	CREATE TABLE match_events (EventID INT PRIMARY KEY, MatchID INT...)	0 row(s) affected	0.048 sec
14	02:10:24	CREATE TABLE standings (StandingID INT PRIMARY KEY, TournamentID INT...)	0 row(s) affected	0.046 sec
15	02:10:25	ALTER TABLE players ADD KitNumber INT DEFAULT 10	0 row(s) affected Records: 0	0.051 sec

DML Outputs:

#	Time	Action	Message	Duration
16	02:12:01	INSERT INTO cities VALUES (1,'Karachi','Sindh'), (2,'Lahore','Punjab')...	5 row(s) affected Records: 5 Duplicates: 0	0.012 sec
17	02:12:02	INSERT INTO venues VALUES (1,'Seaview Futsal Arena'...), (2,'Model Town'...)	5 row(s) affected Records: 5 Duplicates: 0	0.015 sec
18	02:12:03	INSERT INTO users VALUES (1,'Furqan Amir'...), (2,'Hamza Malik'...)	5 row(s) affected Records: 5 Duplicates: 0	0.014 sec
19	02:12:04	INSERT INTO tournaments VALUES (1,'Karachi Futsal Premier League'...)	5 row(s) affected Records: 5 Duplicates: 0	0.019 sec
20	02:12:05	INSERT INTO teams VALUES (1,'Clifton Strikers FC'...), (2,'Lyallpur'...)	5 row(s) affected Records: 5 Duplicates: 0	0.016 sec
21	02:12:06	INSERT INTO players VALUES (101,'Ahmed Raza'...), (102,'Babar Shah'...)	5 row(s) affected Records: 5 Duplicates: 0	0.018 sec
22	02:12:07	INSERT INTO registrations VALUES (1,1,1,'APPROVED'...), (2,1,4...)	5 row(s) affected Records: 5 Duplicates: 0	0.014 sec
23	02:12:08	INSERT INTO payments VALUES (1,1,15000.00,'JAZZCASH'...)	5 row(s) affected Records: 5 Duplicates: 0	0.015 sec
24	02:12:09	INSERT INTO referees VALUES (1,'Tariq Lutfi','NATIONAL'...)	5 row(s) affected Records: 5 Duplicates: 0	0.013 sec
25	02:12:10	INSERT INTO matches VALUES (1,1,1,1,4,1,1,'2026-05-02 18:00:00'...)	5 row(s) affected Records: 5 Duplicates: 0	0.017 sec
26	02:12:11	INSERT INTO match_events VALUES (1,1,1,101,'GOAL',12)...	5 row(s) affected Records: 5 Duplicates: 0	0.015 sec
27	02:12:12	INSERT INTO standings VALUES (1,1,1,3,2,1,0,9,5,4,7)...	5 row(s) affected Records: 5 Duplicates: 0	0.016 sec
28	02:12:13	UPDATE venues SET Capacity = 600 WHERE VenueID = 2	1 row(s) affected Rows matched: 1 Changed: 1	0.008 sec
29	02:12:14	UPDATE players SET KitNumber = 11 WHERE PlayerID = 102	1 row(s) affected Rows matched: 1 Changed: 1	0.007 sec
30	02:12:15	UPDATE payments SET Status = 'VERIFIED' WHERE PaymentID = 5	1 row(s) affected Rows matched: 1 Changed: 1	0.008 sec
31	02:12:16	DELETE FROM match_events WHERE EventID = 5	1 row(s) affected	0.006 sec

QUERIES Output:

• SELECT * :

```
SELECT * FROM cities;
```

Result Grid — Filter Rows: Filter...

#	CityID	Name	Province
▶ 1	1	Karachi	Sindh
2	2	Lahore	Punjab
3	3	Islamabad	ICT
4	4	Rawalpindi	Punjab
5	5	Peshawar	KPK

```
SELECT * FROM venues;
```

Result Grid — Filter Rows:

#	VenueID	Name	Address	CityID	SurfaceType	Capacity	HasFloodlights
▶ 1	1	Seaview Futsal Arena	Clifton Block 4	1	artificial_turf	350	1
2	2	Model Town Sports Complex	Model Town Block J	2	indoor_wooden	600	1
3	3	F-6 Multi-Sport Turf	Super Market F-6	3	artificial_turf	250	1
4	4	DHA Phase 5 Cage Ground	DHA Phase 5	2	artificial_turf	400	1
5	5	Korangi Sports Arena	Sector 31 Korangi	1	sports_court	200	0

```
SELECT * FROM tournaments;
```

Result Grid — Filter Rows:

#	TournamentID	Name	Format	Status	StartDate	EndDate	EntryFeePKR	PrizePoolPKR
▶ 1	1	Karachi Futsal Premier League 2026	LEAGUE	ONGOING	2026-05-01	2026-05-20	15000.00	250000.00
2	2	Lahore Champions Super Cup	KNOCKOUT	REGISTRATION_OPEN	2026-06-10	2026-06-18	20000.00	350000.00
3	3	Margalla Capital Futsal Cup	LEAGUE	DRAFT	2026-07-01	2026-07-15	12000.00	180000.00
4	4	Rawalpindi Night Futsal Trophy	KNOCKOUT	DRAFT	2026-08-05	2026-08-12	10000.00	150000.00
5	5	Sindh Youth Futsal Championship	LEAGUE	COMPLETED	2026-03-01	2026-03-12	12500.00	200000.00

```
SELECT * FROM teams;
```

Result Grid — Filter Rows:

#	TeamID	Name	ShortName	CityID	ManagerID	FoundedYear
▶ 1	1	Clifton Strikers FC	CLS	1	3	2018

2	2	Lyallpur Warriors	LYL	2	4	2019
3	3	Islamabad United Futsal	ISU	3	5	2020
4	4	Karachi Knights	KCK	1	3	2021
5	5	Lahore Badshahs	LBD	2	4	2022

• **WHERE :**

```
SELECT * FROM tournaments WHERE Status = 'ONGOING';
```

Result Grid

#	TournamentID	Name	Format	Status	StartDate	EndDate	EntryFeePKR	PrizePoolPKR
▶ 1	1	Karachi Futsal Premier League 2026	LEAGUE	ONGOING	2026-05-01	2026-05-20	15000.00	250000.00

• **ORDER BY :**

```
SELECT * FROM tournaments ORDER BY PrizePoolPKR DESC;
```

Result Grid

#	TournamentID	Name	Format	Status	EntryFeePKR	PrizePoolPKR
▶ 1	2	Lahore Champions Super Cup	KNOCKOUT	REGISTRATION_OPEN	20000.00	350000.00
2	1	Karachi Futsal Premier League 2026	LEAGUE	ONGOING	15000.00	250000.00
3	5	Sindh Youth Futsal Championship	LEAGUE	COMPLETED	12500.00	200000.00
4	3	Margalla Capital Futsal Cup	LEAGUE	DRAFT	12000.00	180000.00
5	4	Rawalpindi Night Futsal Trophy	KNOCKOUT	DRAFT	10000.00	150000.00

• **DISTINCT :**

```
SELECT DISTINCT SurfaceType FROM venues;
```

Result Grid

#	SurfaceType
▶ 1	artificial_turf
2	indoor_wooden
3	sports_court

• **LIKE :**

```
SELECT * FROM teams WHERE Name LIKE '%Strikers%';
```

Result Grid

#	TeamID	Name	ShortName	CityID	ManagerID	FoundedYear
▶ 1	1	Clifton Strikers FC	CLS	1	3	2018

• **BETWEEN :**

```
SELECT * FROM tournaments WHERE EntryFeePKR BETWEEN 12000 AND 18000;
```

Result Grid

#	TournamentID	Name	Format	StartDate	EndDate	EntryFeePKR	PrizePoolPKR
▶ 1	1	Karachi Futsal Premier League 2026	LEAGUE	2026-05-01	2026-05-20	15000.00	250000.00
2	3	Margalla Capital Futsal Cup	LEAGUE	2026-07-01	2026-07-15	12000.00	180000.00
3	5	Sindh Youth Futsal Championship	LEAGUE	2026-03-01	2026-03-12	12500.00	200000.00

• **JOINs :**

```
SELECT t.Name AS TournamentName, tm.Name AS TeamName, r.Status AS RegistrationStatus, r.RegisteredAt
FROM registrations r
JOIN tournaments t ON r.TournamentID = t.TournamentID
JOIN teams tm ON r.TeamID = tm.TeamID;
```

Result Grid

#	TournamentName	TeamName	RegistrationStatus	RegisteredAt
▶ 1	Karachi Futsal Premier League 2026	Clifton Strikers FC	APPROVED	2026-04-10 10:30:00
2	Karachi Futsal Premier League 2026	Karachi Knights	APPROVED	2026-04-11 14:15:00
3	Lahore Champions Super Cup	Lyallpur Warriors	APPROVED	2026-05-15 11:00:00
4	Lahore Champions Super Cup	Lahore Badshahs	APPROVED	2026-05-16 16:45:00
5	Margalla Capital Futsal Cup	Islamabad United Futsal	PENDING	2026-05-20 09:20:00

• **GROUP BY and AGGREGATED FUNCTION:**

```
SELECT t.Name AS TeamName, SUM(s.GoalsFor) AS TotalGoalsScored
FROM standings s
JOIN teams t ON s.TeamID = t.TeamID
GROUP BY t.TeamID, t.Name;
```

Result Grid

#	TeamName	TotalGoalsScored
▶ 1	Clifton Strikers FC	9
2	Karachi Knights	5
3	Lyallpur Warriors	0
4	Lahore Badshahs	0
5	Islamabad United Futsal	0

• **HAVING and AGGREGATED FUNCTION :**

```
SELECT t.Name AS TeamName, SUM(s.GoalsFor) AS TotalGoalsScored
FROM standings s
JOIN teams t ON s.TeamID = t.TeamID
GROUP BY t.TeamID, t.Name
HAVING SUM(s.GoalsFor) > 5;
```

Result Grid

#	TeamName	TotalGoalsScored
▶ 1	Clifton Strikers FC	9

• **Subquery :**

```
SELECT Name, PrizePoolPKR
FROM tournaments
WHERE PrizePoolPKR > (SELECT AVG(PrizePoolPKR) FROM tournaments);
```

Result Grid

#	Name	PrizePoolPKR

▶ 1	Lahore Champions Super Cup	350000.00
2	Karachi Futsal Premier League 2026	250000.00

- **Multi-table joins :**

```
SELECT m.MatchID, t.Name AS Tournament, ht.Name AS HomeTeam, at.Name AS AwayTeam,
       v.Name AS Venue, m.HomeScore, m.AwayScore, m.Status
FROM matches m
JOIN tournaments t ON m.TournamentID = t.TournamentID
JOIN teams ht ON m.HomeTeamID = ht.TeamID
JOIN teams at ON m.AwayTeamID = at.TeamID
JOIN venues v ON m.VenueID = v.VenueID;
```

Result Grid

#	MatchID	Tournament	HomeTeam	AwayTeam	Venue	Score	Status
▶ 1	1	Karachi Futsal Premier League 2026	Clifton Strikers	Karachi Knights	Seaview Futsal Arena	4 - 2	COMPLETED
2	2	Karachi Futsal Premier League 2026	Karachi Knights	Clifton Strikers	Seaview Futsal Arena	1 - 3	COMPLETED
3	3	Lahore Champions Super Cup	Lyallpur Warriors	Lahore Badshahs	Model Town Sports	0 - 0	SCHEDULED
4	4	Karachi Futsal Premier League 2026	Clifton Strikers	Karachi Knights	Korangi Sports Arena	2 - 2	COMPLETED
5	5	Lahore Champions Super Cup	Lahore Badshahs	Lyallpur Warriors	DHA Phase 5 Cage	0 - 0	SCHEDULED

PHASE 3 – DATABASE SECURITY (DCL)

PLAYGRID — Sports Tournament Management System

Database security is implemented to protect the PLAYGRID Tournament Management System from unauthorized tampering and score adjustments. Two dedicated non-root users are created with different access levels. The administrator is granted full permissions across all tables, while the team manager user is given limited read and registration permissions. One privilege is explicitly revoked to demonstrate role-based access control (RBAC) and database security.

SQL Script:

```
USE PlaygridDB;

/* Create Users */
CREATE USER 'playgrid_admin'@'localhost' IDENTIFIED BY 'admin123';
CREATE USER 'team_manager'@'localhost' IDENTIFIED BY 'manager123';

/* Grant Privileges */
GRANT SELECT, INSERT, UPDATE, DELETE
ON PlaygridDB.*
TO 'playgrid_admin'@'localhost';

GRANT SELECT
ON PlaygridDB.*
TO 'team_manager'@'localhost';

GRANT INSERT
ON PlaygridDB.players
TO 'team_manager'@'localhost';

FLUSH PRIVILEGES;

/* Revoke Privilege */
REVOKE INSERT
ON PlaygridDB.players
FROM 'team_manager'@'localhost';

FLUSH PRIVILEGES;

/* Successful Operations (Executed as playgrid_admin) */
SELECT * FROM tournaments WHERE TournamentID = 1;

INSERT INTO players VALUES
(106, 'Zain Ul Abideen', '42201-9988776-1', '2002-05-18', 'DEFENDER', 1, 5);

UPDATE venues SET Capacity = 650 WHERE VenueID = 1;

DELETE FROM players WHERE PlayerID = 106;

/* Failed Operation (Execute as team_manager) */
INSERT INTO tournaments VALUES
(6, 'Unauthorized Hacker Tournament', 'KNOCKOUT', 'DRAFT', '2026-09-01', '2026-09-10', 50000.00,
1000000.00, 1, 3);
```

Screenshots / Action Output Logs:

User Creation & Privilege Allocation Output:

#	Time	Action	Message	Duration
1	02:22:01	CREATE USER 'playgrid_admin'@'localhost' IDENTIFIED BY 'admin123'	0 row(s) affected	0.003 sec
2	02:22:02	CREATE USER 'team_manager'@'localhost' IDENTIFIED BY 'manager123'	0 row(s) affected	0.002 sec

3	02:22:04	GRANT SELECT, INSERT, UPDATE, DELETE ON PlaygridDB.* TO 'playgrid_admin'@'localhost'	0 row(s) affected	0.001 sec
4	02:22:05	GRANT SELECT ON PlaygridDB.* TO 'team_manager'@'localhost'	0 row(s) affected	0.001 sec
5	02:22:06	GRANT INSERT ON PlaygridDB.players TO 'team_manager'@'localhost'	0 row(s) affected	0.001 sec
6	02:22:08	FLUSH PRIVILEGES	0 row(s) affected	0.004 sec
7	02:22:09	REVOKE INSERT ON PlaygridDB.players FROM 'team_manager'@'localhost'	0 row(s) affected	0.002 sec
8	02:22:10	FLUSH PRIVILEGES	0 row(s) affected	0.003 sec

Successful Operations (Executed as playgrid_admin):

Result Grid

#	TournamentID	Name	Format	Status	EntryFeePKR	PrizePoolPKR
▶ 1	1	Karachi Futsal Premier League 2026	LEAGUE	ONGOING	15000.00	250000.00

#	Time	Action	Message
9	02:24:10	SELECT * FROM tournaments WHERE TournamentID = 1	1 row(s) returned (0.000 sec)
10	02:24:15	INSERT INTO players VALUES (106, 'Zain Ul Abideen', '42201-9988776-1', '2002-05-18', 'DEFENDER', 1, 5)	1 row(s) affected (0.008 sec)
11	02:24:20	UPDATE venues SET Capacity = 650 WHERE VenueID = 1	1 row(s) affected Rows matched: 1 Changed: 1
12	02:24:25	DELETE FROM players WHERE PlayerID = 106	1 row(s) affected (0.006 sec)

Failed Operation (Executed as team_manager):

#	Time	Action	Message
13	02:26:02	INSERT INTO tournaments VALUES (6, 'Unauthorized Hacker Tournament', 'KNOCKOUT', 'DRAFT', '2026-09-01', '2026-09-10', 50000.00, 1000000.00, 1, 3)	**Error Code: 1142. INSERT command denied to user 'team_manager'@'localhost' for table 'tournaments'**

Security Explanation

Database security is important because it protects tournament records and financial prize money data from unauthorized access and misuse. In this system, different users are given different permissions according to their operational roles. The admin user can insert, update, delete, and view data, while the team manager user has limited access restricted to viewing schedules and registering team players. Revoking privileges prevents users from performing unauthorized actions such as creating fake tournaments, modifying match outcomes, or altering prize pool values. This improves data safety, privacy, and integrity in the PLAYGRID database.